

Task I: Given below is an extract taken from the “Literature Review” section of a research paper. Read it and answer the questions that follow.

Dust storms are one of the most dangerous natural hazards and rank only after extreme heat/cold and floods for causing the most deaths and injuries [1]. An increasing number of dust storm events have caused an increase in the reports of the loss of lives and property damages since 1995 (Figure 1). Death and injuries are usually caused by car accidents, because of the reduced visibility during dust storm events [2]. A dust storm forecasting system would be able to alert the public of an approaching dust storm and can help people make safer plans before the potential event. That being said a dust storm forecasting system would be very helpful to enable the public to better prepare for such deadly events and to help reduce potential property damages [2], [3].

Dust storm events pose significant annual variability, for example dust storms usually occur during May and June and from September to November in southwest United States [3],[4]. Typically, most dust storms last from several hours to 1 day. Figure 2 shows the reported yearly frequency and percentage of the total time of dust storm in the United States [4]. It is estimated that the total time of dust storms was generally less than 90 hours and only taking less than 1% of 1 year, if assumed that each dust storm lasts an average of 2 hours. Therefore, a forecasting system for such events would require different computing and access requirements during different times of a year and even different hours within a day [5]. During dust storm events, massive concurrent user accesses to high-resolution dust storm forecasting are expected. The computing platform supporting such an emergency response system should be able to scale up enough computing resources to conduct relevant science, application, and education for the public to make smarter decision for saving more lives and reducing assets loss [5]. After the emergency response, the access to such information will be down and computing resources should be leveraged for other science, application, and education purposes.

Cloud computing is such a computing platform that its computing capabilities can be rapidly and elastically provisioned, in some cases automatically, to scale up and rapidly released to scale down. For different sizes of dust storm forecasting (sizes are determined by the spatial domain, resolution, and temporal scope), different amounts of computing resources are required [5],[6]. For example, 10 KM spatial resolution requires one CPU core, while 4 KM requires at least 16 CPU cores to complete the computing within 2 hours [Figure

3,5]. Therefore, different amounts of computing resources should be allocated for different problem sizes [6]. This also causes disruptive computing requirements that can be best handled by elastic and on demand cloud computing platforms [7]. The emergency response systems, such as dust storm forecasting, require elastic large-scale computing pool [6, 7]. On the other hand, with the development of cloud computing technologies, for example virtualization, and the advancement of cloud infrastructure, cloud computing platforms are ready

to support data and computing-intensive applications. For example, Amazon EC2 offers cluster instances with 10Gbps network connection [8]. Each cluster instance has minimum 23Gbytes memory, a clock speed of 2.93

GHz, and two quad-core processors. Such hardware and network configurations are far better than grid computing environments with heterogeneous computing resources and Wide Area Network (WAN) connection, and even better than most private homogenous high performance computing (HPC) cluster configurations. This makes cloud computing a new advantageous computing paradigm to resolve scientific problems traditionally requiring a special high-performance cluster [9]. The traditional cluster environment requires typically several months and usually hundreds of hours of labor to procure, install, and configure new infrastructure [9]. Traditional processes will involve multiple personnel with different technical backgrounds to ensure the success of the configuration. Oftentimes, greater efforts and costs are required to maintain the infrastructure while the utilization of the infrastructure is only 10-15% of its full capacity [9]. Therefore, cloud computing would have a great impact on parallel scientific HPC applications in the near future by facilitating the deployment process while increasing the rate of utilization of computing resource [15].

1) Based on the above review of literature, which research problem can the researcher logically

derive? a. This study aims to further advance existing grid computing environments used for the detection of dust storms.

b. This study tests the feasibility of using cloud computing to support dust storm forecasting based on Amazon EC2 cloud computing platform. It focuses on costs and capacity assessment. c. This study aims to further advance existing grid computing environments used for the detection of dust storms by employing cloud computing. It tests the feasibility of such a system.

2) What line of argument has the writer followed to present his conceptualization/understanding of literature?

a. Need for dust storm forecasting system---challenges associated with dust storm forecasting--- strengths of existing methods used for forecasting---potential of cloud computing for forecasting

b. Need for dust storm forecasting system---challenges associated with dust storm forecasting--- complex computing requirements---potential of cloud computing for forecasting

c. Need for dust storm forecasting system---challenges associated with dust storm forecasting--- strengths of existing methods used for forecasting---potential of combining cloud computing with existing methods

3) Identify the statements as true or false based on the above review of literature.

a. Total dust storm time is generally estimated to be 90 hours all over the world.

False

b. Dust storm forecasting requires powerful computing resources. **True**

c. One of the strengths of cloud computing is its ability to address unexpected and changing computing requirements. **True**

d. Grid computing environments are a better option due to heterogeneous computing resources and Wide Area Network. **False**

e. Traditional processes deliver up to 20% of their full capacity. **False**

4) In list format, write all the information borrowed by the writer from source [6].

• Different sizes of dust storm forecasting require different amounts of computing resources.
• Example: 10 km spatial resolution requires one CPU core.
• 4 km spatial resolution requires at least 16 CPU cores.
• Emergency response systems like dust storm forecasting require elastic large-scale computing pools.

5) Name any two challenges associated with traditional approaches to forecasting dust storms?

• Long time required to procure, install and configure infrastructure.
• High maintenance cost with low resource utilization (only about 10–15%).

6) Name any two reasons for using cloud computing to predict dust storms?

• It can scale computing resources up or down according to demand.
• It supports large-scale, data-intensive applications efficiently.